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JAN-A-434

31 DECEMBER 1946

JOINT ARMY-NAVY SPECIFICATION

ARMOR, STEEL; PLATE, ROLLED: NON-MAGNETIC (5/32 TO 1–1/16 IN., INCL.) AIRCRAFT-TYPE

This specification was approved by the War Department and the Navy Department for use of the procurement services of the Army and the Navy and supersedes the following specification:

Army-Navy Ordnance Specification
AN–OS–3
15 Sept. 1943

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes; Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

U. S. ARMY SPECIFICATIONS

3–201—Primer, Zinc Yellow.
94–40645—Marking; Shipment (Domestic and Export).[1]
100–2—Standard Specification for Marking Shipments by Contractors.[2]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[3]

FEDERAL SPECIFICATION

QQ–M–151—Metals; General Specification for Inspection Of.

[1] Applicable only to purchases of Army Air Forces.
[2] Applicable only to Army purchases other than those of Army Air Forces.
[3] Applicable only to Navy purchases.

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A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[4]

B. TYPE AND CLASS

B–1. This specification covers one type and class of homogeneous armor made from plates of rolled steel having extremely low magnetic properties as hereinafter specified. Non-magnetic homogeneous armor shall have uniform chemical and physical properties throughout and shall be low enough in magnetic permeability to exhibit for practical purposes inappreciable magnetic properties. (See par. H–1.) This specification covers a range of thicknesses from 5/32 to 1–1/16 inches, inclusive.

C. MATERIAL AND WORKMANSHIP

C–1. *Material and processing.*—The chemical composition of the steel, the heat treatment, and the methods of fabrication will be left entirely to the decision of the armor contractor. No operation which may alter the ballistic characteristics of the armor shall be employed after the ballistic test plate representing a group or lot has been selected.

C–2. *Discard.*—Sufficient discard shall be taken from each ingot used in the manufacture of non-magnetic homogeneous armor to insure freedom of injurious piping, undue segregation, or any other defect which will interfere with the purpose for which the ingot is intended.

C–3. *Defects.*—All plates shall be free of scale, seams, laminations, cracks, blisters, pits, hard or soft spots, or any other defects which are detrimental to the strength or ballistic quality of the armor.

C–4. *Forming.*—The manufacturer shall make all openings and shall bend, form, shape, and drill or punch all plates in accordance with detailed drawings furnished by the prime contractor as approved by the bureau or agency concerned. However, cold-shearing and cold-punching will be permitted on plates of ½-inch thickness or less only upon specific approval of the bureau or agency concerned, provided the temperature of the plate at time of shearing or punching is above 60° F.

C–5. *Local heating or welding.*—No local heating or welding shall be permitted on any plate after heat treatment, except when specifically stated on the drawings or with the approval of the bureau or agency concerned.

D. GENERAL REQUIREMENTS

D–1. See section E.

[4] Applicable only to Navy purchases.

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E. DETAIL REQUIREMENTS

E–1. *Base requirements.*—

E–1a. *Thickness.*—Non-magnetic homogeneous armor shall be furnished in the thickness specified in the contract or order. (See par. H–2.)

E–1b. *Chemical composition.*—Composition of the plates shall conform to the chemical limits established by the contractor in the qualification tests of his product. (See par. F–5.)

E–2. *Soundness, toughness, and ductility.*—The soundness, toughness, and ductility shall be considered acceptable if, as the result of impacts with the designated projectiles, the plates meet the requirements described in paragraph F–4.

E–3. *Hardness.*—The hardness of all plates within a group or lot, including ballistic test plates, shall be within a hardness range of 40 Brinell hardness numbers. The hardness readings taken on any one plate shall not vary by more than 25 Brinell hardness numbers. (See par. F–3f.)

E–4. *Magnetic permeability.*—The magnetic permeability shall be less than 1.08 (ratio) in a magnetic field varying from 0 to 200 oersteds.

E–5. *Ballistic requirements.*—Plates shall withstand the ballistic penetration and shock tests described in paragraph F–4. (See tables I and II.)

E–6. *Fabrication dimensions and tolerances.*—Plates shall be fabricated to the dimensional requirements of the applicable drawings. Unless otherwise specified, all holes and openings shall be machined. Plates shall be capable of being assembled as required by the applicable drawings without the necessity of forcing or jacking other than that which can be done by bolts, jigs, and fixtures normally used in erecting the structures. A template shall be provided by the prime contractor to insure proper fit and interchangeability of the armor. The permissible tolerances for nominal thickness of mill plates shall be as follows:

Nominal thickness	As-rolled width	Tolerance
Inches	Inches	*Inch*
½ and less	36 or less	± 0.015
Greater than ½	36 or less	± .020
All thicknesses	Greater than 36	± .020

E–7. *Information required of the contractor.*

E–7a. *Army.*—The contractor shall furnish to the inspector an executed form approved by the War Department (see par. H–4) for each lot of armor procured by the War Department. Tensile data obtained from the tension test shall be included on the form. This information shall be furnished the inspector prior to forwarding of the ballistic test plates to the proving ground.

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E–7b. *Navy.*—For the purpose of supplying the Bureau of Aeronautics and the Naval Proving Ground with information on which to base ballistic test schedules and workload estimates, the contractor, immediately upon receipt of an order covering material under this specification, shall complete and forward via the Naval inspector a copy of form A (see appendix I) to the Bureau of Aeronautics and a copy to the Naval Proving Ground. Complete information relative to the manufacture of the group represented by the ballistic plates on form B (see appendix II) shall be received at the Bureau of Aeronautics and the Naval Proving Ground before the ballistic tests are conducted.

E–8. *Identification and marking of plates.*—Identification marks and records shall be such as to insure positive identification of all plates including ballistic test plates with the mill heat from which they were produced. Unless otherwise specified in the contract or order, each production plate shall be legibly marked by steel stamping in an area approximately 3 inches square, as indicated on the detail drawing furnished by the prime contractor, with the following information in the specific order indicated:

(a) Manufacturer's (armor contractor's) symbol.
(b) The part number prefixed with the symbol (#).
(c) Mill heat number (may be coded).
(d) Group or lot number.
(e) Serial number.

In addition, the armor contractor shall furnish to the inspector or to the contracting officer a key to the identification marks used on all production and test plates prior to inspection and tests.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of group or lot.*—A group or lot shall consist of all plates of the same nominal thickness from the same mill heat, which have had the same heat treatment, and are submitted at the same time, provided that the total group or lot weight of finished plates does not exceed 20,000 pounds.

F–2. *Sampling.*—

F–2a. *Samples for ballistic tests.*—

F–2a(1). *Army.*—

F–2a(1)a. For contracts under cognizance of the War Department, one flat plate, size 36 by 36 inches, shall be selected from each lot by the inspector for the ballistic tests described in paragraph F–4. If the lot does not contain a flat plate of the size specified herein, two flat plates with dimensions not less than 18 by 36 inches will be selected to represent the lot. Deviation from the minimum size of test plate herein listed will not be permitted unless specifically acceptable to the testing agency concerned.

F–2a(1)b. Whenever possible, test plates will be selected by the inspector after production plates of the lot have been heat-treated. However, upon notification to the inspector, and upon prior approval of the Chief of Ordnance, the test plates may be heat-treated

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prior to the heat treatment of the lot they represent, provided such test plates are processed with production plates of other lots, and the lot which they represent later subjected to the same heat treatment.

F–2a(1)c. In addition to the ballistic test plates, such test plates shall be provided as may be required by the procuring agency or by the inspector.

F–2a(1)d. All test plates submitted shall have a 2½-inch diameter lifting hole with the center 3 inches from the edge and at the middle of one side. In the case of plates which are not square, the hole shall be on the long side.

F–2a(2). *Navy.*—For contracts under cognizance of the Navy Department, after final heat-treatment of all plates in the group, the Naval inspector shall select samples for the ballistic tests in accordance with paragraphs F–2a(2)a to F–2a(2)f, inclusive. As far as practicable the group shall be constituted so as to include large plates having a minimum test area of 12 by 18 inches free of fabricated holes, and small plates of less than 12 by 18 inches free of fabricated holes.

F–2a(2)a. If all plates in the group are greater than 18 by 30 inches, one plate shall be selected.

F–2a(2)b. If all plates in the group are less than 18 by 30 inches but greater than 12 by 18 inches, two plates shall be selected.

F–2a(2)c. If all plates in the group are less than 12 by 18 inches, but greater than 10 by 12 inches, four plates shall be selected.

F–2a(2)d. If all plates in the group are less than 10 by 12 inches, two ballistic test plates 18 by 30 inches from the same mill heat shall be processed with the production plates and one 18- by 30-inch sample test plate together with three production plates shall be selected.

F–2a(2)e. If the group contains large and small plates and the large plates are greater than 18 by 30 inches, one large plate and two small plates shall be selected.

F–2a(2)f. If the group contains large and small plates and the largest plates are less than 18 by 30 inches but greater than 12 by 18 inches, two large and two small plates shall be selected.

F–2a(3). *Marking of ballistic samples.*—All sample plates submitted for ballistic testing shall be legibly painted or stenciled with the contractor's name, contract or order number, heat number or group or lot number, plate number (serially as heat-treated), numbers of applicable specifications, and average weight per square foot (gage). In addition, each plate shall be marked with the inspector's stamp.

F–2b. *Samples for physical tests.*—When required by the bureau or agency concerned, two tension samples, one longitudinal and the other transverse of the plate at least 2T (T = thickness of plate) from the edge of plate shall be taken to obtain the tension test information required on the War Department form (see par. H–4)

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and the Bureau of Aeronautics Form B (see appendix II) in accordance with the requirements of Federal Specification QQ–M–151, and shall represent the group or lot.

F–2c. *Samples for magnetic tests.*—A coupon 10″ x 1″ x T (T = thickness of plate) shall be cut from one edge of each ballistic test plate after final heat treatment. These coupons shall be furnished to the inspector for forwarding to the place of test designated by the bureau or agency concerned.

F–2d. *Thickness determination (test plate).*—The thickness of the test plate as required in the forms described in paragraph E–7 shall be the average of four thickness measurements, one at the center of each side and 2 inches from the edge of the plate. Each measurement shall be read to the nearest .001 inch.

F–3. *Inspection and tests.*—

F–3a. *Purpose of inspection and tests.*—Inspection and tests under this specification shall be for the purpose of

(a) Qualification. (See par. F–5.)
(b) Acceptance of individual groups or lots.

F–3b. *Place of inspection.*—Inspection and tests, other than ballistic, shall be made at the place of manufacture. However, the right is reserved to make inspection and tests at other places at the discretion of the bureau or agency concerned.

F–3c. *Methods of inspection and tests.*—Inspection and tests shall be made in accordance with the requirements of Federal Specification QQ–M–151 unless otherwise specified herein.

F–3d. *Visual and dimensional inspection.*—All finished plates shall be given a thorough visual inspection. The inspector shall measure sufficient plates to assure that the group or lot conforms to the thickness and other dimensional requirements of the applicable drawings. All plates with injurious defects shall be subject to rejection.

F–3e. *Inspection of plate edges.*—Machined and ground edges of all plates shall be carefully examined for incipient cracks, laminations, inclusions, or ruptures without the aid of either a magnetic inspection method or an optical instrument having more than five-power magnification. Any visible cracks, laminations, inclusions, or ruptures shall be cause for rejection of the plate.

F–3f. *Hardness.*—Several Brinell hardness impressions shall be made on each plate in such localities as will permit the general characteristics of the plate to be determined by this method. Before Brinell hardness impressions are made on any part of the plates, any surface scale and decarburization shall be removed from the surface of the plate at that part. The hardness test shall be made with a standard stationary Brinell hardness-testing machine, using a 10-mm. ball and 3,000-kg. load or with a portable Brinell hardness-testing machine previously calibrated for the range involved.

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(The use of a stationary machine is preferred.) Any plate having a Brinell hardness outside the range declared by the armor contractor for that group or lot shall be subject to rejection. When smaller plates are cut from a larger heat-treated plate, hardness readings will not be required on the smaller plates provided the hardness of the larger plate is satisfactory. Brinell hardness tests shall be made in accordance with the requirements of Federal Specification QQ–M–151 and the ball shall be checked frequently for compliance with these requirements. Complete records of Brinell hardness for each group or lot of plates shall be maintained by the contractor. These records shall be submitted to the inspector.

F–3g. *Government analysis.*—At the option of the inspector, a Government chemical analysis may be made on any plate. The sample shall be taken by drilling completely through the plate at any point. Surface chips shall be discarded and the remaining chips shall be thoroughly mixed before analysis.

F-4. Ballistic tests.—

F-4a. *Ballistic definitions.*—In the case of an impact with an armor-piercing projectile at normal obliquity, a penetration shall be called complete when the projectile or projectile core or a major part thereof passes completely through the plate. (Penetration may be determined from the appearance of the impact hole or by the final position of the projectile fragments or both.) The decision of the officer conducting the test shall be final in determining whether or not a given penetration is complete. Any penetration that is not complete is partial (incomplete). The ballistic plate penetration limit (Navy limit) shall be the average of the velocities of, (1) the lowest velocity impact resulting in a complete penetration and (2) the impact with the next lower velocity, disregarding partial (incomplete) penetration resulting in fractured cores stuck in the plate, provided the velocities of these two impacts do not differ by more than 50 foot seconds. (See par. H–5.)

F–4b. *Ballistic test location.*—

F–4b(1). *Army.*—For non-magnetic homogeneous armor purchased under cognizance of the War Department, the ballistic tests shall be conducted at a place designated by the Chief of Ordnance. Ballistic test plates completely identified (see par. E–8) in accordance with the requirements of paragraph F–2a(1) shall be shipped at the contractor's expense to the place designated. (See par. H–4.)

F–4b(2). *Navy.*—For non-magnetic homogeneous armor purchased under cognizance of the Navy Department, the ballistic test plates selected by the Bureau of Aeronautics or its representative shall be shipped at the contractor's expense to the Naval Proving Ground, Dahlgren, Va. (See par. F–2a(2).)

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F–4c. *Penetration and P.T.P. tests.*—

F–4c(1). *Firing procedure.*—The ballistic tests shall consist of a penetration test with caliber-.30 and/or-.50 A.P. projectiles fired as single shots with measured velocities, and any or all of the following tests at the discretion of the officer conducting the tests.

F–4c(1)a. A penetration-thru-plate test with caliber-.30 and/or -.50 A.P. projectiles fired in bursts.

F–4c(1)b. *Additional rounds.*—At the option of the bureau or agency concerned, additional rounds for information may be fired on any test plate after completion of acceptance tests.

F–4c(2). *Number of test impacts.*—At least five impacts with caliber-.30 and/or-.50 projectiles shall be fired in these tests.

F–4d. *Proofing procedure.*—Projectiles, obliquities, and velocities for the ballistic tests shall be as specified in table I. For plates in the range 5/16 to 9/16inch thickness, the officer conducting the test will determine whether caliber-.30 or -.50 projectiles or both shall be used. In the ballistic tests, any round shall be disregarded which, in the opinion of the officer conducting the test, for any reason appears to be unsatisfactory. The penetration-thru-plate tests with caliber-.30 and/or caliber-.50 projectiles shall be fired in one or more bursts at normal obliquity using a striking velocity of 2,450 ± 50 feet per second. The striking velocity of at least one projectile in a burst shall be measured.

TABLE I.—*Penetration tests. Striking velocities.*

Nominal plate thickness	Cal.-30 A.P.M2, normal	Cal.-.50 A.P.M2, normal
Inches	*Feet per second*	*Feet per second*
0.156—5/32	950	
0.188—3/16	1060	
0.250—1/4	1345	
0.313—5/16	1600	1060
0.375—3/8	1820	1230
0.438—7/16	2000	1400
0.500—1/2	2150	1555
0.563—9/16	2290	1705
0.625—5/8		1810
0.688—11/16		1925
0.750—3/4		2030
0.875—7/8		2215
1.000—1		2375
1.063—1-1/16		2455

F–4e. *Interpretation of ballistic tests.*—

F–4e(1). *Failing performance with A.P. projectiles.*—When the test plate is struck by an armor-piercing projectile at normal obliquity at a velocity equal to or less than the tabulated velocity (see table I) plus 15 feet per second, the plate shall have failed if any one or more of the following occurs:

F–4e(1)a. *Penetration.*—A complete penetration providing the impact meets the limitations of paragraph F–4e(2).

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F–4e(1)b. *Shock.*—A crack greater in length than two calibers of the projectiles used, including all cracking during the test or within 24 hours thereafter, providing the impact meets the limitations of paragraph F–4e(2).

F–4e(1)c. *Projectile-thru-plate.*—A hole, the maximum diameter of which, including spall on the back of the plate, exceeds the dimensions shown in table II, providing the impact meets the limitations of paragraph F–4e(2).

TABLE II.—*Projectile-thru-plate test.*

Nominal plate thickness	Projectile	Maximum diameter
Inches	*Caliber*	*Inches*
5/32	.30 A.P. M2	0.7
3/8	.30 A.P. M2	.7
7/16	.30 A.P. M2	.8
1/2	.30 A.P. M2	.8
3/8	.50 A.P. M2	1.0
1/2	.50 A.P. M2	1.0
5/8	.50 A.P. M2	1.2
3/4	.50 A.P. M2	1.2
1	.50 A.P. M2	1.4
1-1/16	.50 A.P. M2	1.4

F–4e(2). *Nonfailing performance.*—At distances measured from center to center of impacts, from center of impact to edge of a fabricated hole, or from center of impact to edge of plate, impacts shall not be cause for failure under the following conditions.

F–4e(2)a. Any impact within three calibers of the projectile used from the edge of the plate or the edge of any fabricated hole.

F–4e(2)b. Any caliber-.30 impact within three calibers of a former caliber-.30 or caliber-.50 impact.

F–4e(2)c. Any caliber-.50 impact within 1.5 inches of former caliber-.30 impact, or caliber-.50 impact.

F–4e(3). *Interpolated thicknesses.*—For thicknesses, other than nominal and not listed in tables I and II, velocities shown in table I and limiting dimensions (maximum diameters) of exit holes shown in table II, shall be obtained by straightline interpolation between the tabulated values.

F–4f. *Rejection.*—Firing at any plate may be stopped whenever, in the opinion of the officer conducting the test, the plate has demonstrated its capacity or incapacity to withstand the full test. If unsuccessful, or if the group or lot to which it belongs is not accepted, each such plate will be returned to the contractor at his expense within 30 days after test. In case of rejection, the contractor shall replace the group or lot within 2 months of the receipt of notice of such rejection.

F–4g. *Retest.*—If the original ballistic test plate fails to pass the ballistic test, the following procedures shall apply.

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F-4g(1). *Retest without reheat treatment.*—Immediately upon notification of the failure of the original ballistic test plate, the inspector, upon the request of the manufacturer, shall select and submit test plates from the failed lot or group, triple the number originally submitted in accordance with the requirements of paragraph F-2a. These plates shall be stamped with the inspector's stamp, and the original group or lot number suffixed by the letter "W". If all additional plates pass the ballistic tests, the group or lot shall be ballistically accepted. If any of these additional plates fail the ballistic tests, the group or lot fails and retest of the group or lot after reheat treatment is authorized.

F-4g(2). *Retest after reheat treatment.*—Immediately upon notification of the failure of the retest plates or of the original test plate, the manufacturer may reheat-treat the entire ballistic group or lot. After reheat treatment, the inspector shall select and submit as test plates from this reheat-treated group or lot, double the number originally submitted in accordance with the requirements of paragraph F-2a. These plates shall be stamped with the inspector's stamp, the original group or lot number suffixed by the letter "R" and also the letter "W" if the group or lot was previously retested without reheat treatment. If all plates pass the ballistic test, the group or lot is acceptable. If any one of these plates fails the ballistic test, the group or lot shall be rejected. The armor in this group or lot shall not be submitted for acceptance again at any time under this specification. If a group or lot of armor is rejected by either the War or Navy Departments under this specification, it shall be considered as rejected by both. No group or lot of armor rejected by either the War or Navy Departments shall subsequently be presented to the other for acceptance.

F-5. *Qualification.*—

F-5a. *Submission of test plates.*—In order to demonstrate the manufacturer's ability to produce non-magnetic homogeneous rolled steel armor by production methods, he shall submit at least four plates 36 by 36 inches of each thickness he proposes to supply. However, if the manufacturer submits and qualifies any nominal thickness listed in table I, he shall be considered as being qualified for the following additional thicknesses.

For nominal plate thickness qualified	Additional thicknesses automatically qualified
Inch	*Inch*
5/32, 3/16, 1/4, 5/16, 3/8, 7/16, 1/2, 9/16	±1/32 from thickness qualified
5/8	+ 1/16, —1/32 from thickness qualified
3/4, 7/8, 1, 1-1/16	±1/16 from thickness qualified

F-5b. *Requirements for qualification.*—The test plates shall conform to all of the requirements of this specification, including flatness to a tolerance acceptable for good commercial practice. All test plates shall pass any or all of the penetration and P.T.P. tests as prescribed in this specification.

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F–5c. *Requalification.*—When a supplier's product shows excessive rejections, he may be removed from the qualification list and, before further production, shall be requalified by passing the tests described in paragraph F–5a. A state of excessive rejection shall be established only by a consensus of all of the procuring agencies interested in the supplier involved.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Rust preventive treatment.*—Unless otherwise specified, all plates of non-magnetic homogeneous armor shall be coated with a rust preventive conforming to the requirements of U. S. Army Specification 8–201.

G–2. *Packing.*—

G–2a. *For domestic shipment.*—Plates of non-magnetic homogeneous armor treated as described in paragraph G–1 shall be shipped in bundles or packed in standard commercial containers so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

G–2b. *For overseas shipment.*—Unless otherwise specified, plates of non-magnetic homogeneous armor treated as described in paragraph G–1 shall be packed in nailed wooden boxes or cleated plywood boxes conforming to the requirements of Joint Army-Navy specifications JAN–P–106 and JAN–P–105, respectively, except that waterproof case liners will not be required. For cleated plywood boxes, no water-resistant or toxic-treated plywood will be required. Each box shall contain multiples of complete armor sets and the gross weight shall not exceed 200 pounds.

G–3. *Marking for shipment.*—

G–3a. *Bundles.*—Unless otherwise specified, bundles of armor plates shall have one outside plate legibly marked with the following information completed:

ARMOR: .....
PART NO: .....
STOCK NO: .....
ORDER NO: .....
MANUFACTURER: .....
MANUFACTURER'S IDENTIFICATION SYMBOL: .....
QUANTITY: .....
GROSS WEIGHT: .....

G–3b. *Shipping containers.*—In addition to any special marking required by contract or order, shipments for the Army, except Army Air Forces, shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook. For Army Air Forces procurement, each exterior shipping container shall be marked in accordance with the requirements of U. S. Army Specification 94–40645.

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H. NOTES

H–1. *Intended Use.*—The material covered by this specification is intended primarily for use in aircraft as protective armor for installation near equipment, the functioning of which may be affected by external magnetic fields. Also, this material may be used elsewhere for protection of equipment under similar conditions.

H–2. Requests, requisitions, schedules, and contracts or orders should contain the following information:

(a) Title, number, and date of the specification.
(b) Thickness required. (See par. E–1a.)
(c) Whether domestic or overseas shipment is required. (See par. G–2.)

H–3. *Quality.*—In the manufacture of the armor described herein, the contractor should make every endeavor, and incur all necessary expenses, to furnish the very best armor that can be produced at the time of manufacture. The tests set forth in these specifications are to be regarded merely as minimum limits of quality which shall be exceeded if practicable and it should be incumbent upon the contractor to make every effort during the course of manufacture to improve the ballistic quality of the armor furnished.

H–4. *Ordnance Department, War Department, Ballistic Test Summary Sheet.*—For purposes of collecting all data pertinent to ballistic testing, it is required by paragraphs E–7a and F–4b(1) that the supplier of armor furnish all data requested on approved form. This is known as form CAS–2 and is filled out in accordance with instruction issued for use of that form. The proving ground or range will not test the plate ballistically until form CAS–2 is received, properly filled out. (See par. F–4b(1).) These forms with instructions are obtainable at the local Army Ordnance district office.

H–5. *Ballistic criterion.*—All material procured under this specification will be ballistically tested to give the Navy limits. (See par. F–4a.)

H–6. Copies of U. S. Army Specification 94–40645 may be obtained upon application to the Commanding General, Air Materiel Command, Wright Field, Dayton, Ohio.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and United States Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

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H–8. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–9. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

H–10. Copies of this Joint Army-Navy specification (required for Navy purchases) may be obtained only upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D. C. When requesting, state the purpose for which the specification is desired.

ARMY: OA.

NAVY: AC*G*OSHS.

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APPENDIX I

FORM A

From: .....
To: The Chief of the Bureau of Aeronautics.
The Commanding Officer,
Naval Proving Ground, Dahlgren, Virginia.
Subject: Light Armor—Submission of Data.

1. The following data are submitted regarding the manufacture of light armor requiring ballistic test at the Naval Proving Ground:

(*a*) Armor Manufacturer (name) .....
Located at (city) .....
in District of Inspector of Naval Material located at (city) .....
(*b*) Armor Contract or Purchase Order No.....
Dated .....
(*c*) Prime Contractor .....
.....
(*d*) Prime Contract No.....dated.....
(*e*) Type and Purpose.....
(*f*) Specifications No.....Table No.....
(*g*) Total Armor on This Order.....lbs.
(*h*) Estimated No. Ballistic Test Plates to be submitted.
No..... Gage.....
No..... Gage.....
(*i*) Estimated Schedule of Submission of Test Plates.
.....
.....
[s] .....

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APPENDIX II

FORM B

From: .....
To: The Chief of the Bureau of Aeronautics,
The Commanding Officer,
Naval Proving Ground, Dahlgren, Virginia.
Subject: Light Armor—Ballistic Test Plate Data.

1. Plate No.....Representing Group No.....
   Submitted by (Mfr.).....
2. Reasons for Test.....
3. If for acceptance, Order & Contract No.....
4. For Test under Specification No.....Table.....
5. Dimensions of Plate.....Average Gage.....
6. Estimated Wgt.....lbs. Actual Wgt.....lbs.
7. Mill Heat No.....Made by.....
8. Physical Properties (after final heat treatment).
   Yield Point.....PSI. Tensile .....PSI.
   Elongation in 2 inches.....%. Reduction in area.....%
   Percent Depth of Case.....
9. Chemical Composition: Furnace; Electric...Open Hearth...

	C	Si	Mn	S	P	Ni	Cr	Mo	V
(Face)....	....	....	....	....	....	....	....	....	....
(Back)....	....	....	....	....	....	....	....	....	....

10. Heat treatment given this plate and group.
    Annealing Temp..... Time at temp. .....
    Normalizing Temp..... Time at temp. .....
    Hardening Temp. ..... Time at temp. .....
    Quenching Medium ..... Temp. ....
    Drawing Temp. ..... Time at temp. .....
11. Finished Brinell Hardness: Face..... Back.....
12. Depth of Brinell Polish: Face..... Back.....
13. Remarks .....

[s] .....
Inspector

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